TriAlign

Progress in comparing clinical trial eligibility criteria

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Project objective

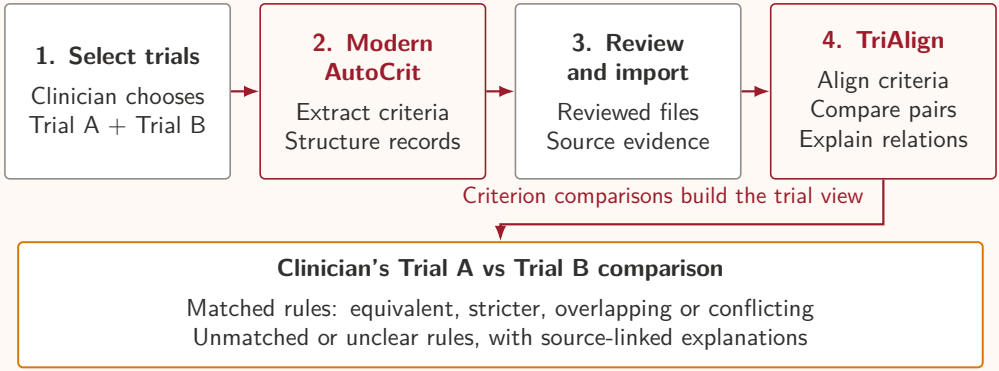
Help a clinician compare the eligibility requirements of Trial A and Trial B.

1. Align potentially corresponding criteria across the trials.
2. Explain relationships and differences using source evidence.
3. Identify criteria needing further review and preserve unresolved comparisons.

Illustrative case: inclusion age ≥ 18 versus inclusion age < 18.
Shared words conceal disjoint allowed populations.

Scope: trial comparison through criterion alignment, without patient screening.

From extracted criteria to trial comparison



Today: Modern AutoCrit runs separately. The TriAlign demo starts from reviewed criteria.

Current dataset: already extracted and reviewed

Data source	Public oncology and cardiovascular trial eligibility text
Extraction	Modern AutoCrit produces atomic criteria with structured fields
Manual review	Reviewed by the project owner in `trial100_v1.xlsx`
Frozen corpus	3,537 criteria across 100 trials, stored as Parquet
Evidence retained	Criterion text, parent statements, trial IDs and source context

What the demo loads: select two trials from this frozen corpus, or import other extracted criterion records.

The demo compares existing criteria. Fresh extraction remains in Modern AutoCrit.

What does a clinical relationship mean?

Compare the people allowed by each **selected rule**, within a common frame.

Relationship	Rule A: age	Rule B: age	Interpretation
Equivalent	≥ 18	≥ 18	Same allowed set
A more restrictive	≥ 21	≥ 18	A allows a smaller set
B more restrictive	≥ 18	≥ 21	B allows a smaller set
Partial overlap	18–65	50–80	Shared ages, plus ages unique to each
Contradictory / disjoint	≥ 18	< 18	No shared allowed ages

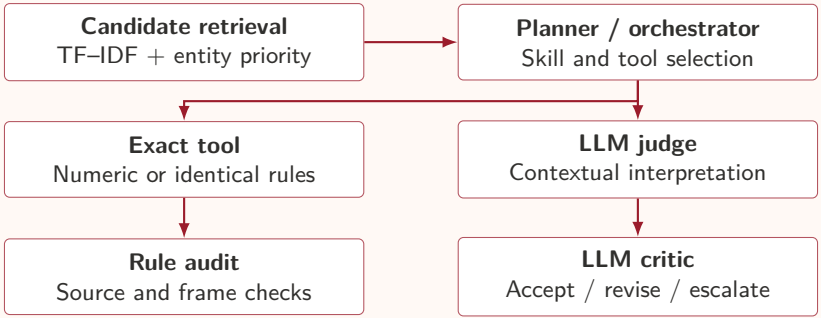
Related, not comparable: related concepts without a supported common frame.

Unrelated: different concepts without a relevant relationship.

Ambiguous: missing or conflicting source meaning prevents a decision.

Age examples are illustrative inclusion rules in years. Role, units, timing and Boolean scope can change the relationship.

How the agentic comparison runs



Outcome: an evidence-supported proposal or an unresolved case for human review.

The orchestrator allows one revision, then finalizes or escalates. Retrieval similarity alone never supplies a clinical label.

Skills, tools and hooks give the agents boundaries

Component	Role in the working system
Skills	Versioned instructions for numeric comparison, clinical context and source problems. The planner selects a skill.
Tools	Inspect the comparison frame, compare supported constraints and collect source citations. Only permitted tools run.
Hooks	Validate inputs and exact source quotes, check limits and cancellation, and gate the final result.
Orchestrator	Coordinates tool calls, judge, critic and revision within pair, call and time limits.

Clinician-facing output: source excerpts, rationale, critic feedback, executed trace and saved reviewer decisions.

Contextual judge and critic use separate prompts with the configured LLM (currently GPT-4o mini). Agreement does not establish clinical correctness.

Retrieval progress and the model still to train

Implemented by Jiatong

- TF–IDF, MPNet and BioLORD retrieval backends.
- Cross-trial ranking with a shared neighbor format.
- Provisional runs on 200 criteria across 100 trials.

Current local demo

- Uses the original TF–IDF encoder as the temporary retriever.
- Exact tools and LLM reasoning assign relationships.
- Dense baseline integration awaits the teammate PR.

Next model step: use reviewed comparisons to train and evaluate a clinically informed encoder and directional relationship component.

Clinical retrieval quality and improvement over the baselines are not yet measured.

Where TriAlign fits among eligibility systems

TriAlign: compare criteria across two trials and explain equivalence, restrictiveness, overlap or uncertainty using source evidence.

System	Primary task	Planned connection to TriAlign
TrialGPT	Patient-to-trial matching	Adapt evidence-based judgments to common criterion pairs.
EligMeta	Evidence synthesis	Compare eligibility alignment and LLM / deterministic execution.
CReSE	Criterion recommendation	Compare embeddings if reproducible artifacts permit.
SatIR / VERDICT	Logical matching and rationales	Study consistency, explicit assumptions and explanation faithfulness.
EC-RAFT	Criterion generation	Adjacent authoring work, outside the direct benchmark.

Planned evaluation: common cases and evidence where tasks align. Compare labels, evidence support, abstention, latency and cost. No external-system results yet.

What the prototype resolves and where it stops

Trial A: NCT05379712 Trial B: NCT02769832

Supported age comparison

A: "$\geq$ 18 years of age"

B: "Age $\geq$ 18 years old, both male and female"

Equivalent in the age frame.

The exact tool compares allowed ages. This does not establish equivalence of the whole trials.

Consent comparison needing review

A: ability to read, understand and sign consent.

B: willingness and ability to sign consent.

No supported final label in the saved run.

The source-quote check rejected the model's citations. The case remains unresolved for review.

These are observed prototype outcomes. Shared consent wording does not establish equivalence of literacy, understanding and willingness.

Current challenges and limitations

- Clinical interpretation can fail on timing, exceptions, negation and parent/child scope.
- Judge and critic can share mistakes. Human review remains necessary.
- No retrieved candidate does not prove that a requirement is absent.
- Independent clinical evaluation, final family-aware splits and calibrated uncertainty remain pending.
- Dense baseline integration and fine-tuning remain ahead.

The UI runs locally. Contextual model calls still need network access and have API costs and latency.

Next steps before the final presentation

Owner	Deliverable
Jonathan	Finish review of development comparisons and source ambiguities. Finalize family-aware evaluation partitions.
Jiatong	Submit the retrieval PR, integrate MPNet/BioLORD and record reproducible model revisions.
Together	Evaluate retrieval and relationship predictions against reviewed labels. Analyze errors and critic behavior.
Together	Improve and rehearse the local interface, including evidence inspection and reviewer corrections.
Together	Train the clinical encoder once labels and evaluation partitions are ready, then compare with frozen baselines.

Longer term: let Modern AutoCrit and TriAlign run separately or together in one tool.

Questions and feedback

Which differences matter most when comparing two trial protocols?

What evidence would help you trust or correct a proposed relationship?

When should the system stop and ask for clinical review?